

Group discussion

Case study



Case 1: Scurvy

Mr.colombus works as a sailor, he has been used for many years to using canned food to obtain his daily dietary requirements. He observed that he as well as his colleagues have swollen gums bleed on brushing their teeth , as well as delayed healing of their wounds on shaving . Moreover, he has observed that his hair is not easily brushed and it looks always spiky like corkscrew.

On arrival to the nearest port , he consulted a doctor who observed that colombus has a weak loose skin , as well as redness and swelling of his gums. When the doctor asked colombus to flex and extend his knees, colombus failed to do so , and the doctor noticed that colombus has swollen knees as well as bloody spots seen over his skin. The blood picture showed HB level 10 gm/dl, platelet count was normal . Prothrombin time and concentration was normal. Bleeding time was prolonged



Inflammation of the gums common in first signs of Scurvy

Q1: What is the most likely diagnosis?

Q2: What is the biochemical bases of the symptoms?

Q3:What would be the probable treatment?

Q4: Do you know another name of Vitamin C?

Q5 MCQs



A1- Our case is a scurvy (vitamin C deficiency).

A2- As regards the bleeding all over the body as well as the delayed healing of wounds , Bruises are due to the defect in collagen synthesis that is necessary for healing as well as the strength of the skin .

*The anemia is due to the deficiency of vitamin C which is essential for absorption of iron from the gut. As it reduces ferric (Fe^{3+}) to Ferrous (Fe^{2+}) that is the absorbable form.

A3- Treatment is mainly dietary by eating citrus fruits as Lemon, orange , and Lorange as well as taking supplementary daily tablets of vitamin C and trying to depend on fresh prepared beverages instead of canned foods.

A4- L-Ascorbic acid

MCQs

1-The diagnosis of this condition is:

- a. Rickets
- b. Pernicious anemia
- c. Scurvy**
- d. Osteomalacia

2- Loosening and falling of teeth, gums is swollen and infected and bleeds easily in deficiency.

- a. Vitamin A
- b. Vitamin C**
- c. Vitamin K
- d. Vitamin B1

3-Which one of the following statements describe vitamin C?

- a. it is derived from fructose
- b. 90% of it, present in the form of dehydroascorbic acid
- c. it is involved in collagen synthesis**
- d. its deficiency causes beriberi

MCQs

4-Which of the following statements describe water-soluble vitamins?

- a- Their absorption is influenced by bile salts
- b-They need carrier protein
- c-They cannot be stored in the body**
- d-They include vitamin A&D

5-Absorption of iron requires which one of the following vitamins?

- a- Vitamin A
- b- Vitamin C**
- c- Vitamin B
- d- Vitamin D

Case 2: Type 2 Diabetes mellitus

A 52-year-old obese male with high socioeconomic standards presents with fatigue, polyuria, polydipsia and unexplained weight loss . Fasting blood glucose is 160 mg/dL and HbA1c is 8.2%.

Q1: What is the diagnosis and key diagnostic criteria?

A: Type 2 Diabetes Mellitus.

Criteria include fasting glucose ≥ 126 mg/dL or HbA1c $\geq 6.5\%$.

Short essay questions on the case

Q2: What are the main risk factors in this patient?

- A: Obesity, age >45 years, sedentary lifestyle,

Q3: What is the initial management?

A: Lifestyle modification (diet, exercise), weight loss, and metformin.

Q4: What are possible complications?

- A: Microvascular (retinopathy, nephropathy, neuropathy) and macrovascular (cardiovascular disease).

Type 2 Diabetes mellitus :Case 2:

A 64-year-old man is presented to his family doctor with complaints of frequent episodes of numbness in his legs, frequent urination and thirst sensation. During a routine history and physical examination, the doctor finds that the patient had a sedentary lifestyle, he is obese, and has hypertension (blood pressure of 200/120 mm Hg). The patient is asked to return to the lab in the fasting state, during which time a blood specimen is obtained and analysed for blood sugar,. The analysis reveals fasting hyperglycemia. The diagnosis is type II diabetes mellitus.

MCQs

1- Diabetes mellitus is diagnosed when fasting blood glucose level is

- a. ≥ 70 mg/dl
- b. ≥ 126 mg/dl
- c. ≥ 141 mg/dl
- d. ≥ 200 mg/dl

2-Normal serum two hours postprandial glucose level is.....

- a. < 70 mg/dl
- b. < 126 mg/dl
- c. ≤ 140 mg/dl
- d. < 200 mg/dl

3- A pateint with fasting blood glucose level of 116 mg/dl is diagnosed as:

- a. normal
- b. Prediabetic
- c. Diabetic
- d. Hypoglycemia

4- Excessive thirst in diabetes is called:

- a. Polyuria
- b. Polydipsia
- c. Polyphagia
- d. Dysuria

MCQs

5- Which cells produce insulin?

- a. Alpha cells of langerhans
- b. **Beta cells of langerhans**
- c. Delta cells
- d. Acinar cells

6-Which complication is acute in diabetes?

- a. Neuropathy
- b. Retinopathy
- c. **Diabetic ketoacidosis**
- d. Nephropathy

7- Diabetic ketoacidosis is more common in:

- a. Type 2 diabetes
- b. **Type 1 diabetes**
- c. Gestational diabetes
- d. Prediabetes

8- Which of the following is a chronic complication?

- a. Hypoglycemia
- b. Ketoacidosis
- c. **Retinopathy**
- d. Hyperglycemia

Case 3: Metabolic acidosis

- Uncontrolled diabetic person was admitted to hospital presented with with rapid breathing, confusion, and fatigue. Analysis of the arterial blood gases showed the following values: pH =7.25, PCO₂=25 mm Hg, HCO₃ =14 mmol/l ,Normal Values: pH 7.35-7.45, P CO₂ [35 – 45], HCO₃ [22 - 28].

MCQs

- **1- The diagnosis of this case is:**
 - a. Metabolic Acidosis
 - b. Metabolic Alkalosis
 - c. Respiratory Acidosis
 - d. Respiratory Alkalosis
- **2-. What is the most likely cause?**
 - a. Diarrhea
 - b. Diabetic ketoacidosis
 - c. Vomiting
 - d. Hyperventilation

MCQs

3- The cause of low PCO₂ levels is:

- a. Partial compensation of respiratory acidosis
- b. Complete compensation of metabolic acidosis
- c. Complete compensation of respiratory acidosis
- d. Partial compensation of metabolic acidosis

• **4- The the most significant buffer system in plasma is**

- (A) Phosphate buffer system
- (B) Carbonic acid-bicarbonate buffer system
- (C) Lactic acid-lactate buffer system
- (D) Protein buffer system

Short Essay Questions

- Q1: Explain the pathophysiology of metabolic acidosis in this case.
- A: Insulin deficiency leads to increased lipolysis and ketone production, causing accumulation of acids and decreased bicarbonate.
- Q2: How does the body compensate?
- A: Respiratory compensation via hyperventilation to decrease CO₂ and raise pH

Short Essay Questions

- Q3: Outline the management.

A: Fluid resuscitation,
insulin therapy,
electrolyte correction,
and treatment of underlying cause.

Case 4: Vitamin D deficiency

Age: infant 18 months. Main Complaint: The parents noticed "bowed legs", box-shaped skull, delayed eruption of teeth and a delay in the child's ability to walk independently.

1. What is the diagnosis of the case?

- a. Ricket
- b. Anemia
- c. Osteomalacia
- d. Beri Beri

MCQs

2. What is the cause of this case?

- a. deficiency of vitamin D
- b. deficiency of vitamin C
- c. deficiency of iron
- d. deficiency of vitamin A

3. The most biologically potent Vitamin D metabolite is

- (A) 25-Hydroxycholecalciferol
- (B) 1,25-Dihydroxycholecalciferol
- (C) 24, 25-Dihydroxycholecalciferol
- (D) 1,24 -hydroxycholecalceferol

4. Deficiency of vitamin D in adults causes

- (A) Anemia
- (B) Tuberculosis of bone
- (C) Hypothyroidism
- (D) Osteomalacia

MCQs

5. vitamin D is all of the following except?

- a. water soluble vitamin
- b. stored in liver
- c. activated in liver and kidney
- d. fat soluble vitamin

6. All the following statements describe vitamin D EXCEPT:

- a- It is derived from glucose
- b- synthesized in skin by UV light
- c-its deficiency cause Osteomalacia
- d-it is needed for calcium absorption from intestine

7. All the following statements describe rickets Except:

- a. It is caused by deficiency of vitamin D in infants
- b. It is characterized by high plasma calcium levels
- c. It is characterized by box shaped skull
- d. There is knock-knee

Short Essay Questions

- Q1. What is the primary role of Vitamin D in bone health, and how does a deficiency lead to rickets or osteomalacia?
- Answer: Vitamin D is essential for the intestinal absorption of calcium and phosphorus. Without sufficient Vitamin D, the body cannot maintain adequate levels of these minerals in the blood. In children, this leads to rickets, causing soft bones and skeletal deformities. In adults, it causes osteomalacia

Short Essay Questions

Q2. Differentiate between Rickets and Osteomalacia

Both conditions involve defective mineralization of the bone matrix.

However, Rickets occurs in children whose epiphyseal plates have not yet closed, leading to skeletal deformities like bow-legs and box skull.

Osteomalacia occurs in adults after the epiphyseal plates have closed. the bone's matrix becomes soft and bones easily fractured.

Short Essay Questions

Q3. Explain the relationship between sunlight exposure and Vitamin D synthesis in the skin.

Answer: The body synthesizes Vitamin D naturally when skin is exposed to ultraviolet B (UVB) radiation from sunlight. UVB rays react with a cholesterol derivative called 7-dehydrocholesterol in the epidermal layers of the skin to produce Vitamin D3 (cholecalciferol).

Factors such as high melanin levels (darker skin), geographical latitude, season, and the use of sunscreen can significantly reduce the efficiency of this synthesis, potentially leading to deficiency.

THANK
YOU